

Surveillance of *Aedes aegypti* in Madeira, Portugal: Insights for Climate-Driven Vector Modelling for Portugal

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BACKGROUND

Aedes aegypti introduction and establishment

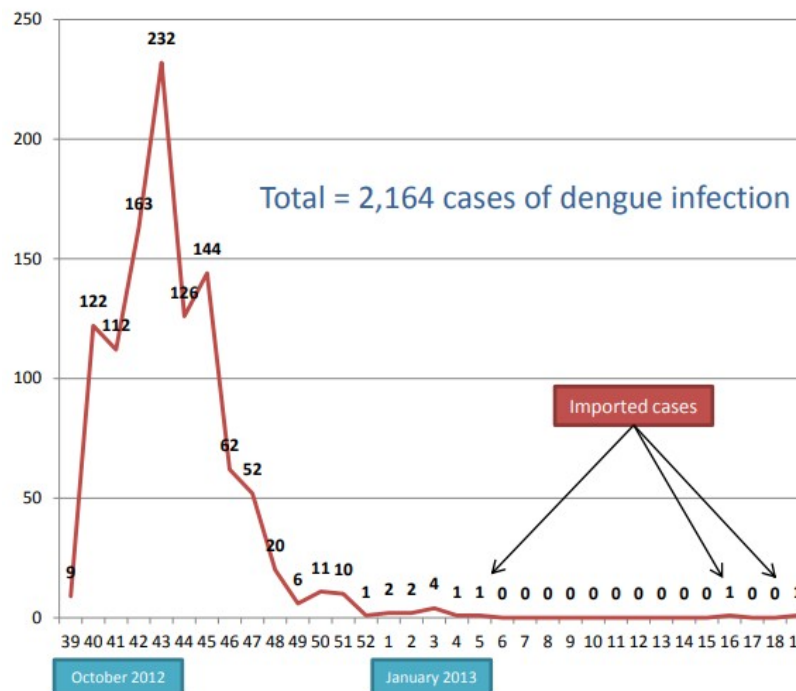
Santa Luzia, Madeira



Dengue outbreak 2012

- Dengue outbreak (DENV-1) with over 2,000 cases and 80 exported cases in 2012

Number of new dengue cases in Madeira/ week



- No deaths or cases of severe dengue have been reported
- All reported cases refer to the resident population of the Island

EU imported cases from Madeira

Portugal (mainland)	11
UK	23
Germany	19
France	3
Sweden	5
Finland	7
Denmark	2
Austria	2
Norway	2
Croatia	1
Slovenia	1
Spain	1
Switzerland	1
TOTAL	78

Source: ECDC

Last case reported on
1 February 2013 from
Finland

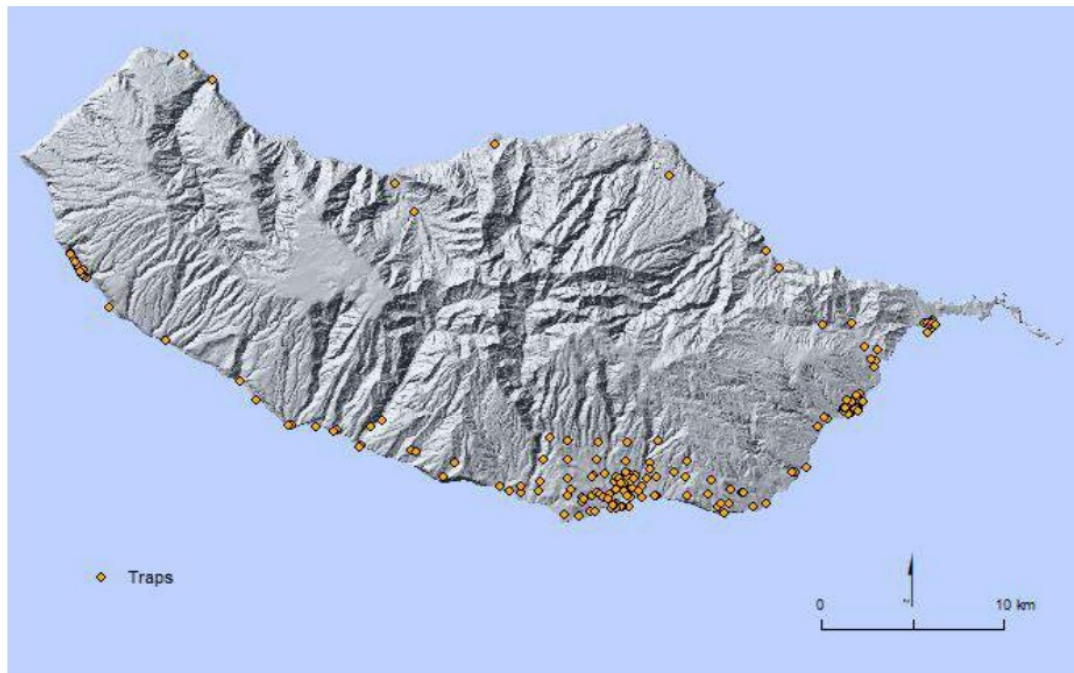
Public health impact:

- First autochthonous dengue outbreak in Europe in recent decades
- Triggered intensified entomological surveillance and control measures

Surveillance Programs

The **Regional Directorate of Health** conducts continuous surveillance monitoring vector populations and disease risk and participating in the National Vector Surveillance Program – **REVIVE**

Spatial distribution of mosquito traps



Urban Presence and Control Efforts

Persistent *Aedes aegypti* presence in Funchal and Câmara de Lobos highlights the need for ongoing monitoring and control



Environmental and Geographic Factors

Madeira's unique climate and geography provide a natural laboratory to study vector population dynamics and environmental impacts

- Temperate mediterranean climate, with low temperature variation – from 18°C in January to 23°C in August
- Annual rainfall close to 700mm





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SURVEILLANCE METHODOLOGY

Monitoring techniques and data collection



- **Mosquito Surveillance Methods**

Ovitrap (N=124) and BG-Sentinel traps (N=11) are used to monitor immature and adult *Aedes aegypti* mosquito populations weekly

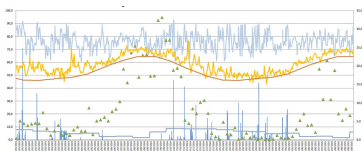


- **Specimen Analysis Process**

Collected mosquito specimens are preserved and sent for species identification and RT-PCR viral testing

- **Data Integration and Application**

Surveillance data is combined with meteorological variables to correlate mosquito activity with environmental factors for public health responses.





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SURVEILLANCE RESULTS

Weekly and accumulated data for 2025

(Week 36: September 1-7)

INDICATOR	WEEKLY (WEEK 36)	ACCUMULATED (2025)
Eggs	3629	48747
Ovitrap Read	124	5713
Ovitrap Positive	40	690
Adult Mosquitoes	6	644
BG-Traps Read	11	870
BG-Traps Positive	4	186

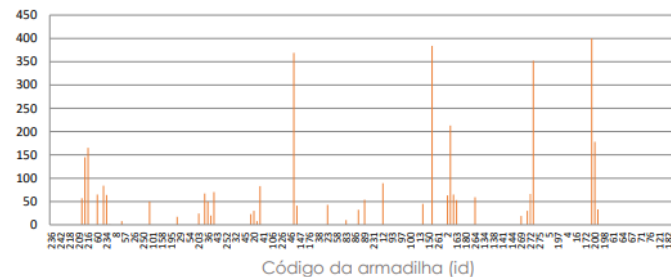
Week 36: 01-07 September 2025 – Ovitrap

OVITRAPS

Weekly results

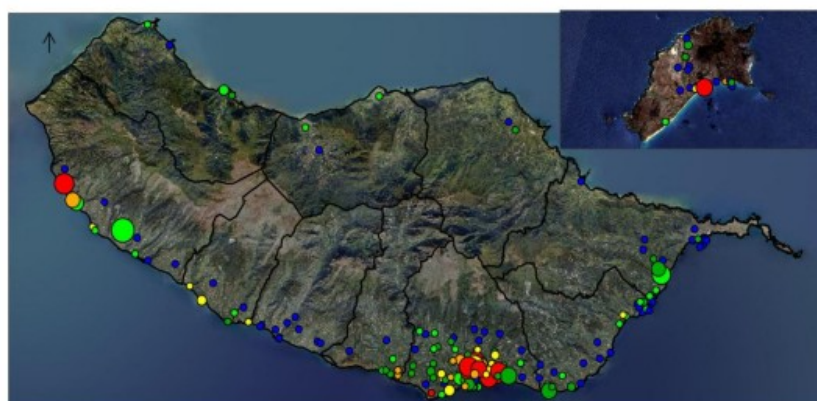


N.º of eggs

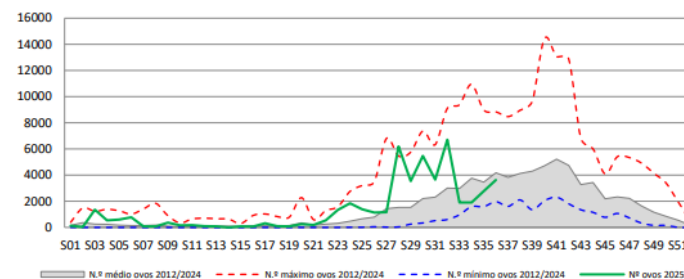


Indicadores	N.º de ovos	N.º de ovitraps c/leitura	N.º de ovitraps pos.
	3629	124	40

Accumulated results



N.º of eggs

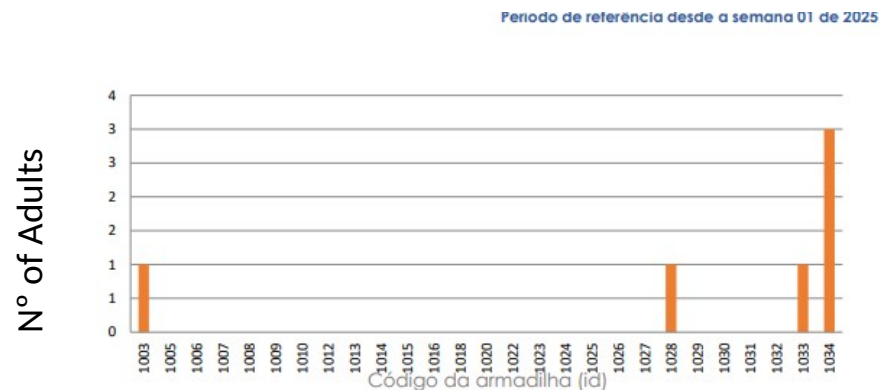
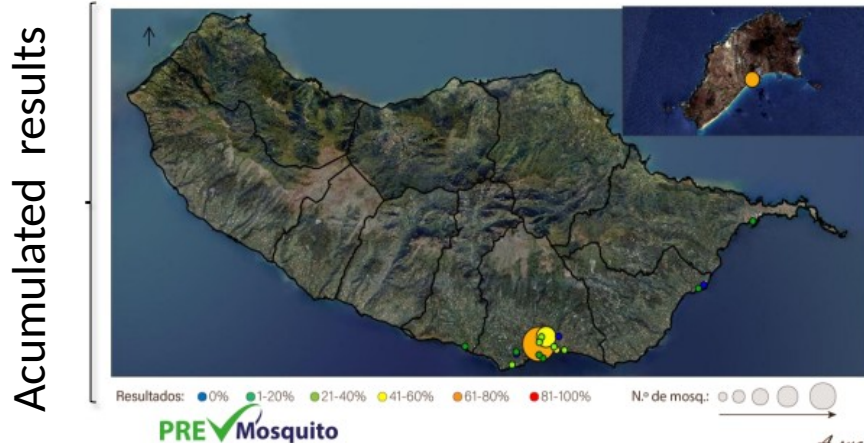


Indicadores	N.º de ovos	N.º de leituras	N.º de positivities
	48747	5713	690

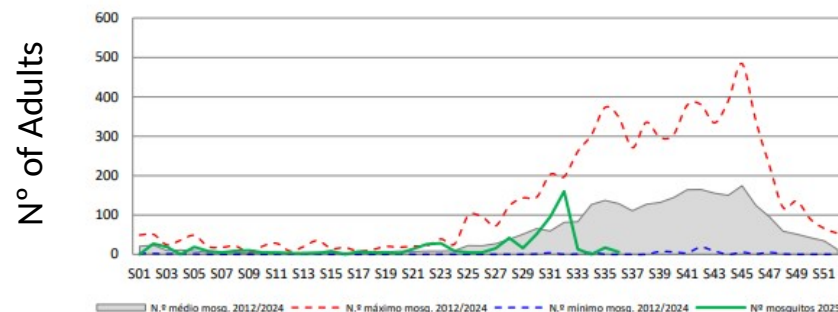
PRE Mosquito

Week 36: 01-07 September 2025 – BGS

BG – Sentinel traps



Indicadores	N.º de mosquitos	N.º de BGtraps c/leitura	N.º de BGtraps pos.
	6	11	4

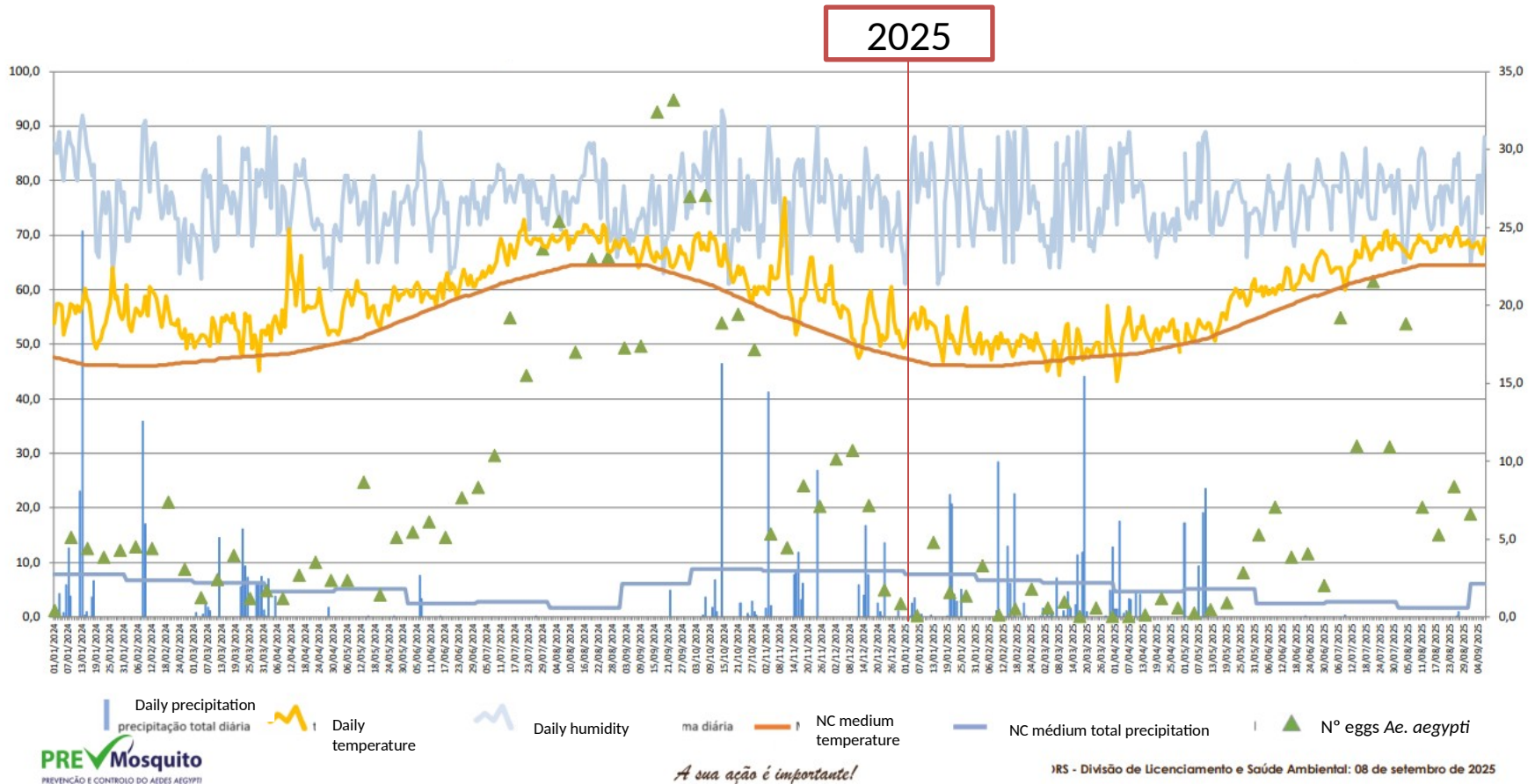


Indicadores	N.º de mosquitos	N.º de leituras	N.º de positivities
	644	870	186

A sua ação é importante!

IRS - Divisão de Licenciamento e Saúde Ambiental: 08 de setembro de 2025

Meteo and climatic data (NC-1971-2000) and entomological data week_1 2024 to week_36 2025

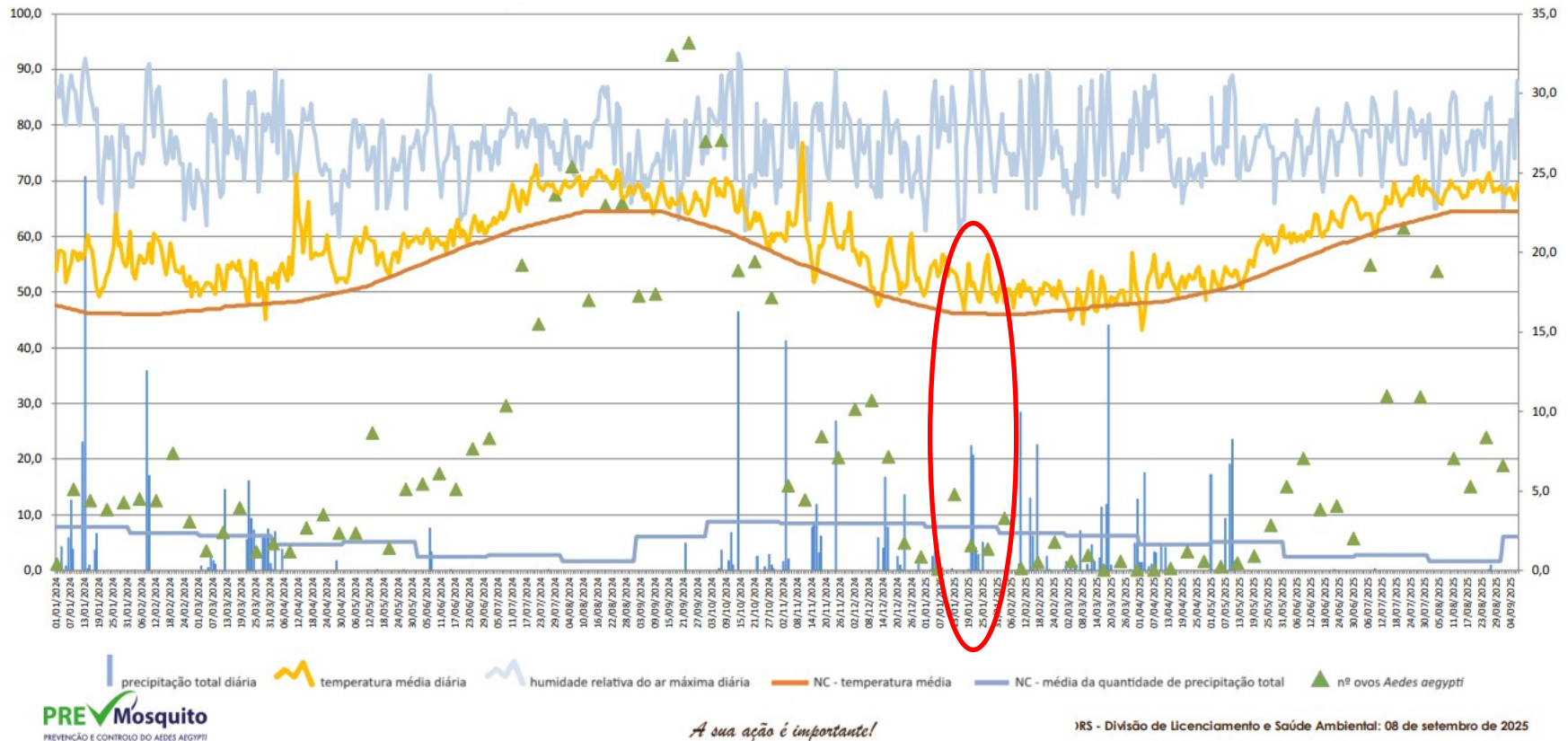




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SURVEILLANCE CASE STUDY

Case Study – DENV-2 Detection in Madeira (Jan 2025)



- Low mosquito abundance and IndexP < 1 in January
- Suggests off-season introduction with limited transmission



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DISCUSSION

Climate correlation

Preliminary analyses of climate impact

- **Climate Variables Influencing Mosquitoes**

Mosquito activity correlates with temperature, humidity and precipitation

- **Seasonal Mosquito Patterns**

Egg and adult mosquito counts peak during warmer, more humid seasons – July - September

- **Data Integration for Analysis**

Combining meteorological and entomological data to quantify climate-vector relationships is essential for **Predictive Models for Public Health:**

- Developing models forecasts mosquito abundance to improve **early warning** and **vector-control strategies**

Take home messages

- **Importance of Surveillance**

Continuous *Aedes aegypti* presence demands sustained surveillance and proactive public health measures in Madeira

- **Data for Climate Modelling**

Under **REVIVE program** data can support climate-driven vector modelling to anticipate future vector-borne disease risks

- **Collaborative Efforts**

Success relies on the collaboration among REVIVE team, DRS Madeira, NIH and stakeholders

- **Informed Public Health Decisions**

Reliable data enables informed decision-making and enhances understanding of vector dynamics within climate change

Plataform No Mosquito

- <https://www.iasaude.pt/Mosquito/>



- School projects
- Citizen Science
- Travelers information
- Entomological panels
- Training and education in Public Health



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